School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Group-6 Intelligent Transportation System Scribe

Scribe (Part 1) : Q1, Q2, Q3

Question 1: Baseline System and Motivation for Randomization

Deterministic Baseline Strategy

The deterministic baseline implemented in Milestone 3 is based on fixed route selection using the A* shortest path algorithm. In this approach, the A* (A-star) algorithm computes a route that minimizes the expected total travel time based on deterministic edge weights. Once the route is selected, the route remains fixed throughout the simulation. This baseline calculates the shortest path by assuming completely fixed, deterministic travel times for each route edge. The travel time for each edge is calculated using fixed formula : $\text{travel\_time} = \frac{\text{length}}{\text{speed\_limit}}$

Implementation of Randomness in System

Implementation of randomization has improved system's performance by modeling uncertainty in travel times and deadlines, allowing the algorithm to evaluate a fixed route across many realistic scenarios and compute reliability (probability of success) instead of just relying on a single deterministic estimate.

In the deterministic A* baseline implemented in previous milestone, a single fixed route is decided based on static edge (segment) costs. The route is fixed, and its travel time is taken as a constant value. As a result, the model ignores real-world uncertainty such as traffic congestion and variability in travel conditions.

In the implemented randomized approach, the route still remains fixed (obtained from A*), but its evaluation is made stochastic using Monte Carlo simulation. Randomness is introduced in the travel time of each edge (segment) and in the deadline. Each edge travel time is sampled from lognormal distribution and deadlines are sampled from normal distribution.

This improves system performance because:

- It considers realistic traffic variability (such as congestion, delays) through stochastic edge travel times instead of relying on deterministic estimate of travel time
- It evaluates the same fixed route under many possible traffic scenarios using Monte Carlo simulation
- It measures reliability using the probability of meeting a (randomized) deadline rather than considering fixed threshold
- The algorithm captures variability in travel times (for e.g., congestion spikes)

Hence, the system is able to estimate expected travel time more accurately, and optimize for reliability rather than just shortest distance. It leads to better real-world performance under uncertainty.

Modifications introduced in Baseline

In the implementation of randomized approach, the route selection step of A* is not modified. The modification is there in the evaluation of the route cost (travel time) and deadline.

The list of modifications is as follows :

- Each edge travel time T_i is modeled as a random variable:

$$T_i \sim \text{LogNormal}(\mu_i, \sigma_i)$$

  This introduces variability due to traffic conditions.

- The algorithm performs multiple simulation runs. In each run:
  - Segment (Edge) travel times are sampled randomly
  - Total route travel time is computed

  This shifts the output from a single value to a distribution

- Unlike deterministic baseline (fixed deadline), modified approach samples deadlines as :

$$D \sim \mathcal{N}(\mu_d, \sigma_d)$$

  which captures variability as per external conditions.

- The modified approach gives these outputs:
  - Mean travel time
  - Standard deviation
  - Confidence intervals

  which allows statistical analysis of performance.

Question 2 : Randomized Algorithm Design

The implemented algorithm combines deterministic A* algorithm with a randomized evaluation framework based on Monte Carlo simulation. The route is first computed using a deterministic method (A* algorithm) and then evaluated under uncertainty by introducing randomness in travel times and deadlines.

Decision Component Involving Randomness

Here, the travel time of each road segment (edge) is treated as a random variable, and the total route travel time is obtained as the sum of these random variables across all segments or edges. Also, the deadline used for comparison is also randomized (denoted by random variable D as discussed in above question). As a result, the decision component that involves randomness is the evaluation of whether the route meets the deadline under varying travel conditions or not. Instead of a fixed outcome, the algorithm determines the likelihood of successfully reaching the destination within the deadline.

For each road segment i, the deterministic travel time is first computed as:

$$t_i = \frac{\text{length}_i}{\text{speed}_i}$$

This value is then scaled using a congestion factor:

$$\mu_i^{(\text{mean})} = t_i \times \text{congestion\_factor}$$

The total route travel time for iteration k is:

$$T_{\text{route}}^{(k)} = \sum_{i=1}^{n} T_i^{(k)}$$

where each $T_i^{(k)}$ is randomly sampled. The decision component involving randomness is:

$$T_{\text{route}}^{(k)} < D^{(k)}$$

which determines whether the route meets the deadline in each simulation run.

Probability Distribution and Sampling Mechanism

This approach mainly uses two probability distributions to model uncertainty. The travel time for each edge is modeled using lognormal distribution, which ensures that all sampled values are positive and captures the right-skewed nature of traffic delays. The parameters of this distribution are derived from the deterministic travel time and adjusted using congestion and variance factors.

Each segment travel time is modeled as:

$$T_i \sim \text{LogNormal}(\mu_i, \sigma_i)$$

The parameters are derived as:

$$\sigma_i^2 = \ln\left(1 + \frac{\text{Var}_i}{\mu_i^2}\right), \quad \mu_i = \ln(\mu_i^{(\text{mean})}) - \frac{\sigma_i^2}{2}$$

where:

$$\text{Var}_i = (\text{variance\_factor} \times \mu_i^{(\text{mean})})^2$$

This transformation ensures that the sampled values have the desired mean and variance.

The deadline is modeled using normal distribution, which captures variability in specific external constraints (for example, user expectations).

$$D \sim \mathcal{N}(\mu_D, \sigma_D)$$

where:

$$\mu_D = 1.4 \times T_{\text{base}}, \quad \sigma_D = 0.15 \times \mu_D$$

Here T_{base} is the deterministic total travel time from A*.

In this algorithm, Monte Carlo simulation is used as sampling mechanism to repeatedly sample from these distributions over multiple iterations. In each iteration, edge travel times are sampled and aggregated to obtain the total route time. This repeated sampling generates a distribution of possible outcomes.

Difference from Deterministic Baseline

In the deterministic baseline, the route is fixed and evaluated using constant travel times derived from edge length and speed. It gives a single travel time estimate as output and does not account for uncertainty.

However, the randomized algorithm evaluates the same fixed route under stochastic conditions. Travel times are no longer fixed as they vary across simulation runs, and the deadline is also treated as a random variable. As a result, the output is no longer a single value but a distribution of travel times (and deadlines).

The decision rule is also modified. Instead of minimizing travel time, the algorithm evaluates the probability that the total travel time is less than the deadline. This shifts the objective from minimizing travel time to assessing the probability of completing the route within the specified deadline under uncertainty.

Question 3 : Probabilistic Reasoning Behind the Algorithm

The randomized approach is considered due to the need of modeling uncertainty in travel times and evaluate route performance under realistic traffic conditions. Instead of relying on deterministic estimate, the randomized approach treats travel time and deadlines as random variables and analyzes their behavior using probabilistic methods.

Importance of Randomness in Addressing Uncertainty

In real-life situations, travel time on each road segment (edge) is uncertain due to congestion and varying traffic conditions. The deterministic estimate

$$t_i = \frac{\text{length}_i}{\text{speed}_i}$$

does not capture this variability.

To address this issue, each segment travel time is modeled as a random variable :

$$T_i \sim \text{LogNormal}(\mu_i, \sigma_i)$$

As a result, the total route time is also a random variable:

$$T_{\text{route}} = \sum_{i=1}^{n} T_i$$

Randomness allows the algorithm to simulate multiple possible traffic conditions and evaluate how the route performs across these scenarios instead of assuming a single fixed outcome.

Probabilistic Assumptions and Models

The randomized algorithm is based on the following probabilistic assumptions:

- Independent Segment Travel Times: T_i (Travel Time) for different segments (or edges) are assumed to be independent random variables. This simplifies the modeling and allows the total travel time to be computed as a sum.
- Lognormal Model for Travel Time: Each T_i follows a lognormal distribution, ensuring:
  - $T_i > 0$ (as travel time cannot be negative or zero)

   - Right-skewed behavior due to possible delays

- Normal Model for Deadline: The deadline is modeled as:

$$D \sim \mathcal{N}(\mu_D, \sigma_D)$$

  capturing variability in external time constraints.

- Monte Carlo Estimation: The probability of meeting the deadline is estimated empirically:

$$P(T_{\text{route}} < D) \approx \frac{1}{N} \sum_{k=1}^{N} \mathbf{1} \left(T_{\text{route}}^{(k)} < D^{(k)} \right)$$

  where N is the number of simulation runs.

Randomized Strategy Improves Performance (Reasons)

The deterministic baseline evaluates the route using a single travel time estimate, which may be inaccurate under variable traffic conditions. On the other hand, the randomized strategy evaluates the route over a distribution of possible outcomes.

By estimating the following probability :

$$P(T_{\text{route}} < D),$$

the algorithm measures the reliability of the route rather than just its expected performance.

This improves overall system's performance because:

- It captures variability and extreme delays which are generally not considered in deterministic models.
- It provides statistical measures such as mean, variance, and confidence intervals.
- It enables decision-making based on reliability (probability of success) rather than considering fixed and deterministic estimate.

End of Submission